Program Outcomes:

Engineering Graduates will be able to:

1. **Engineering knowledge:**
2. **Problem analysis:**
3. **Design/development of solutions:**
4. **Conduct investigations of complex problems:**
5. **Modern tool usage:**
6. **The engineer and society:**
7. **Environment and sustainability:**
8. **Ethics:**
9. **Individual and team work:**
10. **Communication:**.
11. **Project management and finance:**
12. **Life-long learning:**

Program Specific Outcomes (PSO) addressed by the Course:

A graduate of the Civil Engineering Program will be able to

PSO1: Work in Civil Engineering sector which involves various aspects of planning, designing and construction of structures.

PSO2: Undertake higher studies, identify complex problems of Civil Engineering and formulate research thinking.

PSO3: Use comprehended knowledge in the broad domain of Civil Engineering to undertake various competitive examination.

Course Outcomes Mapping

CO	Description	Blooms Level	Mapping with POs & PSOs
CO1	Describe various types of transportation systems and their history of the development	2 - Understand	PO1,2,6,8,12 & PSO1,2
CO2	Discuss various types of highway materials and pavements	2 - Understand	PO1,2,3,12 & PSO1,2,3
CO3	Analyze the pavements by considering various aspects associated with traffic safety measures.	4 - Analysis	PO1,2,3,12 & PSO1,2,3

Topic Learning Outcomes

Sr. No.	TLO	CO	TLO Description	BL
1	TLO1	CO1	Understand significance of various modes of transportation	2 - Understand
2	TLO2		Discuss developments in road construction and Classify roads based on certain criterion	1 - Remember
3	TLO3		Describe factors for selecting road alignments and carrying out road surveys	2 - Understand
4	TLO4	CO2	Interpret knowledge about various materials used for highway construction	2 - Understand
5	TLO5		Describe tests carried out on materials used for highway construction.	1 - Remember
6	TLO6		Introduce the fundamental concepts of highway construction	2 - Understand
7	TLO7		Describe pavements on the basis of structural behavior	2 - Understand
8	TLO8	CO3	Design different road geometrical elements	3 - Apply
9	TLO9		Identify traffic stream characteristics and Design a pre-timed signalized intersection.	3 - Apply
10	TL10		Identify causes of road accidents and measures for road safety.	2 - Understand
11	TL11		Analyze the important features of pavement designing considering various aspects	4 - Analysis

Module 1: Introduction (6 Lectures)

- Importance of various modes of transportation
- Highway Engineering, Road Classification
- Developments in Road Construction, Highway Planning, Alignment and Surveys

Module 2: (6 Lectures)

- Geometric Design- Cross section elements, Sight distances
- Horizontal alignment Vertical alignment, Intersections,
- Construction of Pavements
- Construction and Maintenance of Drainage
- Road Arboriculture

Module 3: (8 Lectures)

- Highway Materials: Soil relevant properties, Various tests,
- Aggregates – strength, hardness, toughness, soundness, durability, shape, specific gravity, water absorption
- Bituminous materials – Bitumen, Tar, and Asphalt – various properties
- Design of Bituminous paving mixes-Marshall stability test

- **Module 4: Traffic Engineering (8 Lectures)**
- Traffic Characteristics, Speed, Journey Time and Delays, Vehicle Volume Counts, Origin and Destination Studies, Analysis and Interpretation of Survey Data, Traffic Operations
- Design of Signals and Rotary intersections, Parking Space Design, Highway Lighting, Planning and Administration, Road Markings, Signs

•**Road Accidents and Safety:** Classification, Causes, Mitigation and Control Measures, Aspects of Safety in Usage of Roads, Type and Design of anti-crash barriers

- Introduction to Intelligent Transport Systems (ITS).

Module 5: Pavement Design (8 Lectures)

- Basic Principles, Methods for different Types of Pavements, Design of flexible pavement using IRC: 37- 2012, Design of rigid pavement using IRC: 58-2011

Other modes of Transport

- Introduction to Railways, Airways, Waterways, Pipeline Transportation, Classification, Requirements, Comparative Studies

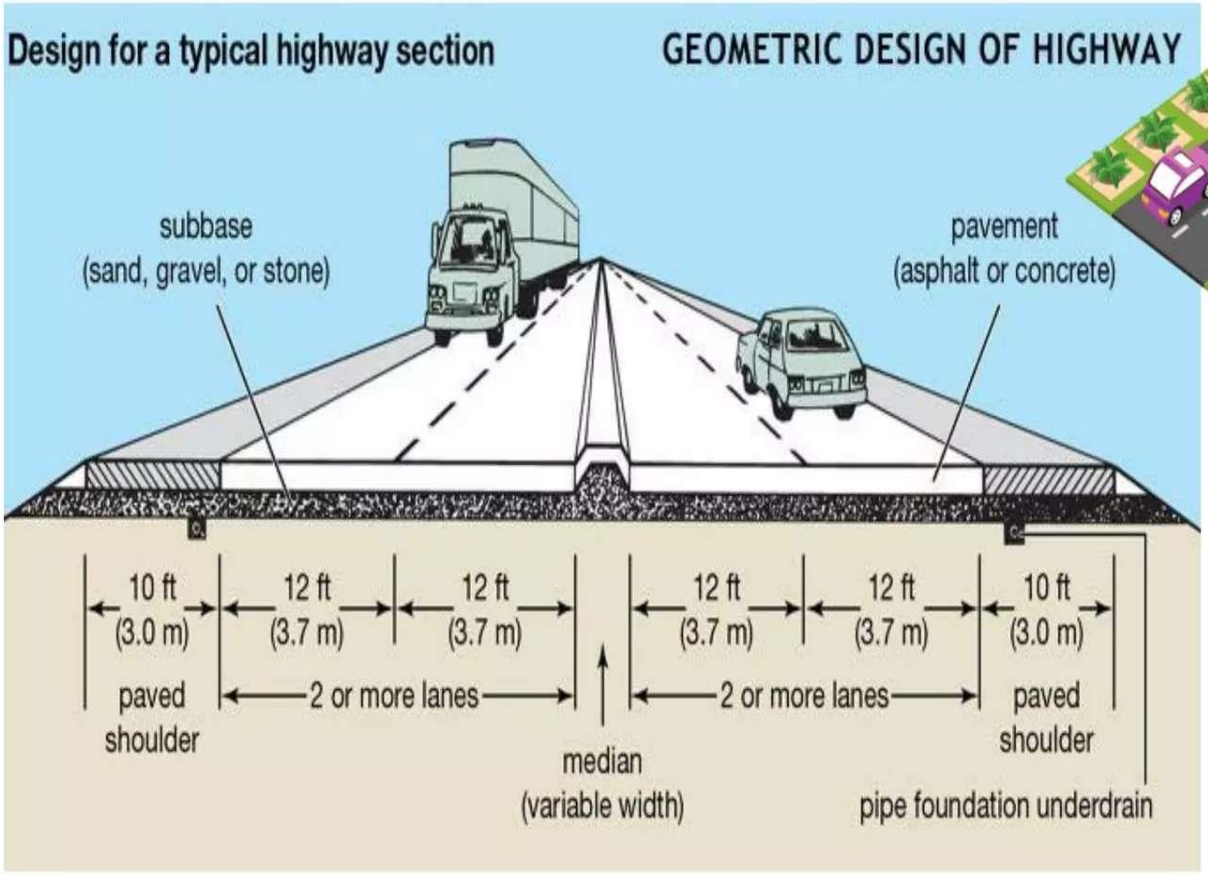
References

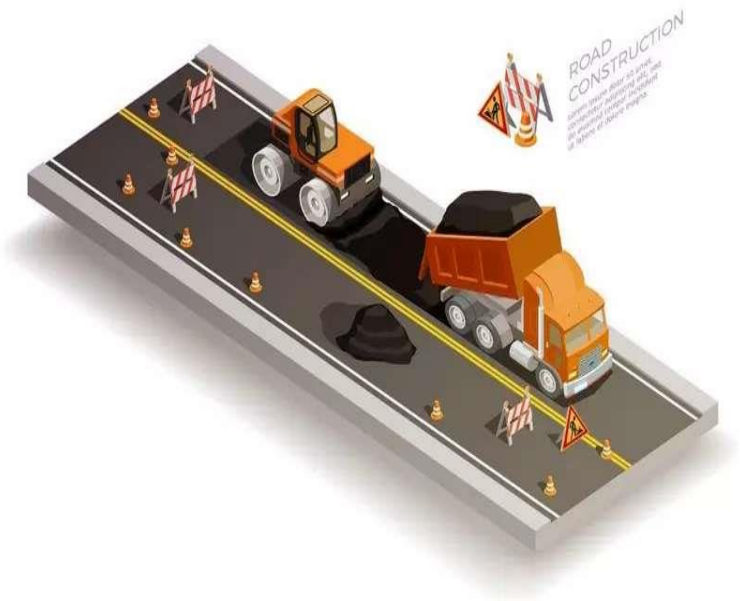
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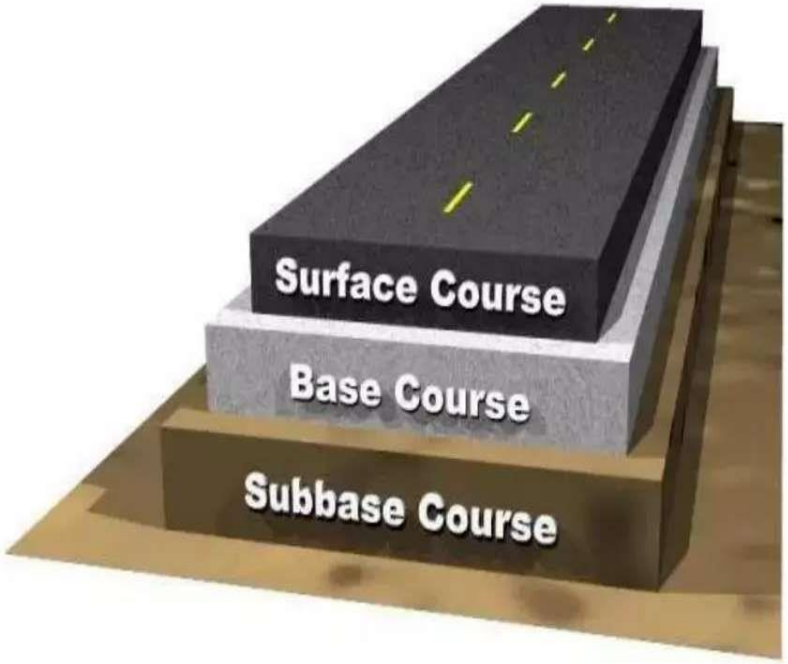


Design for a typical highway section

GEOMETRIC DESIGN OF HIGHWAY




ROAD
CONSTRUCTION


Surface Course
Base Course
Subbase Course











Modes of transport

Land Transport

Moves on land.



Bus



Car

Water Transport

Moves on water.



Ship



Boat

Air Transport

Runs in air.



Helicopter



Aeroplane

Objective of Transportation System:-

- Main objective of a good transportation system is to provide **safe, economical, efficient transportation facility** for the travel of passengers and transportation of goods.

Role of Transportation:

1) Transportation is an index of **economic, social, cultural, industrial & commercial progress of a country.**
2) Transportation is vital for the **economic development** of any region, whether it is **agricultural or industrial product** needs to be transported from production to distribution.
3) Transportation is also required for **carrying raw material for production**.
4) The whole **structure of industry and commerce** rests on the well laid foundation of transport.
5) No **country** or region can ever **flourish** if it lacks adequate transport facilities.
6) Main role of transportation is **to provide safe, economical, efficient** transportation facility for the **travel of passengers and transportation of goods**.

Importance of Transportation System:-

1) **Economic Activity & Transport:** production, supply, consumption, human demand.

2) **Social Effects of Transportation:** Population have always settled along the routes.

3) **Sectionalism:** Cultural exchange, living condition improvement, increase of knowledge, etc.

4) **Concentration of Population in Urban Area:** Employment opportunity & superior facilities.

5) **Aspects of Safety, Law & Order:** emergency, external aggression, etc.

6) **Development in Rural Area:** 75 % population living in rural area, balanced development of city & village.

Classification of Transport:

a) Human portar

b) Animal transport

c) Road transport

d) Rail transport

e) Air transport

1) Human energy
2) Animal energy
3) Petrol & diesel energy
4) Steam energy
5) Electric energy











Different Mode of Transportation:

Classification from Surface Point of View:

1) **Land Transport:** Highways, Railways, Cableways, etc.
2) **Water Transport:** Canal ways, River ways, Ocean ways, Lake ways, etc.
3) **Air Transport:** Airways.

Classification based on Freedom to Move:

1) **One Degree Freedom:** Vehicles are free to move along a line. Ex. Railways.
2) **Two Degree Freedom:** move along a line as well as laterally. Ex. Ship, bats, etc.
3) **Three Degree Freedom:** free to move in any plane. Ex. Aero-planes.

Different Mode of Transportation:

1) **Road Transport:** Transportation by road system is the **only mode which could give maximum flexibility of service from origin to destination**, to one & all. Nearest to people

2) **Rail Transport:** Transportation **could be economical between the railway station** & advantages for passenger & goods for long distance. (as resistance to traction along the track for the movement is much lower)

3) **Water Transport:** Transportation is **most energy efficient but slow.**

4) **Air Transport: Fastest among the four modes**, Comfortable, **energy requirement more, costly** & affected by weather condition.

Characteristics	Railways Transport	Highway Transport
Load handling Capacity.	Heavier loads at high speed.	Low.
Right of entry.	Not free to all.	Free & flexible.
Operational Control.	Signaling, interlocking & block system.	Not required.
Tractive Resistance.	Less (1/6 th of highway)	More.
Gradient.	Minimum.	Steeper Gradient.
Constr./Maintenance.	Higher.	Less.
Origin & Destination.	Starting & Destination points are fixed.	Door to door service.
Length of Haul.	Bulk & Heavy goods is cheaper (Convenient).	Short distance upto 500km is convenient.
Employment.	Less.	Higher.
Hilly Regions.	Not suitable.	Suitable.
Accident rate.	Few.	More.
Net tonnes-kms/vehicle hr.	Higher.	Lower.
Horse Power.	Lesser per tonnes	Higher per tonnes.

Characteristics of Road Transport:

1)Roads are used by passenger car, buses, trucks, 2-3 wheel automobile, cycle, animal drawn vehicle & pedestrian.

2)Requires **lowest initial investment** (O & M) than other mode like railway, airport & harbor's.

3)Offers **complete freedom** to users according to need & convenience.

4)Flexibility in location, direction, speed & timings of travel, which is not available in other modes of transport.

5)Possible to **travel directly** from respective places of origin to the destination.

6)For **short distance** saves time & most convenient.

7)Accidents in other modes are **more severe** & disastrous in spite of various safety measure & strict control.

8)Only mode that offers the **facilities** to the **whole section of society**.

Importance of Roads in India:

1)The **deficiency in the road development** has contributed to the **set back in the growth** of **agricultural, industrial & commercial sectors** in the country. Hence, for **balanced development** of any country, it is essential to provide a well planned road network connecting all the towns and villages.

2) It is necessary to provide **good road links between the villages & market centers**.

3)Overall **economic progress** can be achieved, only if **adequate transport facilities** are made available between villages. district & commercial centers.

4)It is also important to **develop high quality highways, expressways, connecting important** urban, industrial & commercial centers.

5)Road developments generates considerable **employment potential**, which is of additional significance to the developing country.

6)Paved surface in good condition can contribute to **15 to 40 % savings in vehicle operation cost.**

Road Engineering: (Highway Engineering)

1) The branch of transportation engineering **which deals with the design, construction & maintenance of different types of roads** is called **Road Engineering.**

2) The **science & technology dealing** with roads is generally called **Highway Engineering**.

Traffic: a vehicles, cyclist, pedestrian, etc. travelling together on a road constitute the traffic.

Roadway: comprises of carriageway plus shoulder on either side.

Motorway : portion of roadway of an **urban road** reserved for use **only by high speed & power driven vehicles**. This is also referred as expressway or super highway.

Scope of Highway Engineering:

1)Development, planning & location of roads.

2)Highway design.

3)Material required for construction & maintenance.

4)Highway traffic performance & control.

5)Economics, finance & administration.

Development of Road Construction:

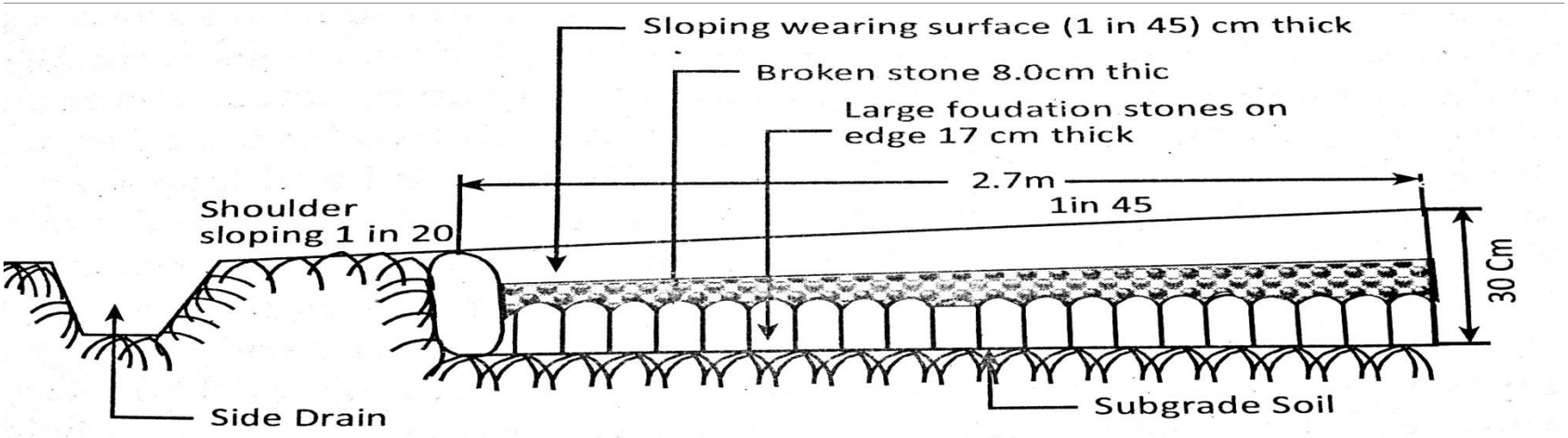
1)The first road was constructed in 1900 B.C. by Assyrian empire as per authentic record.

2)Romans are the pioneers in ancient road construction 312 B.C.

3)Roads were constructed on a large scale only during the period of Roman empire.

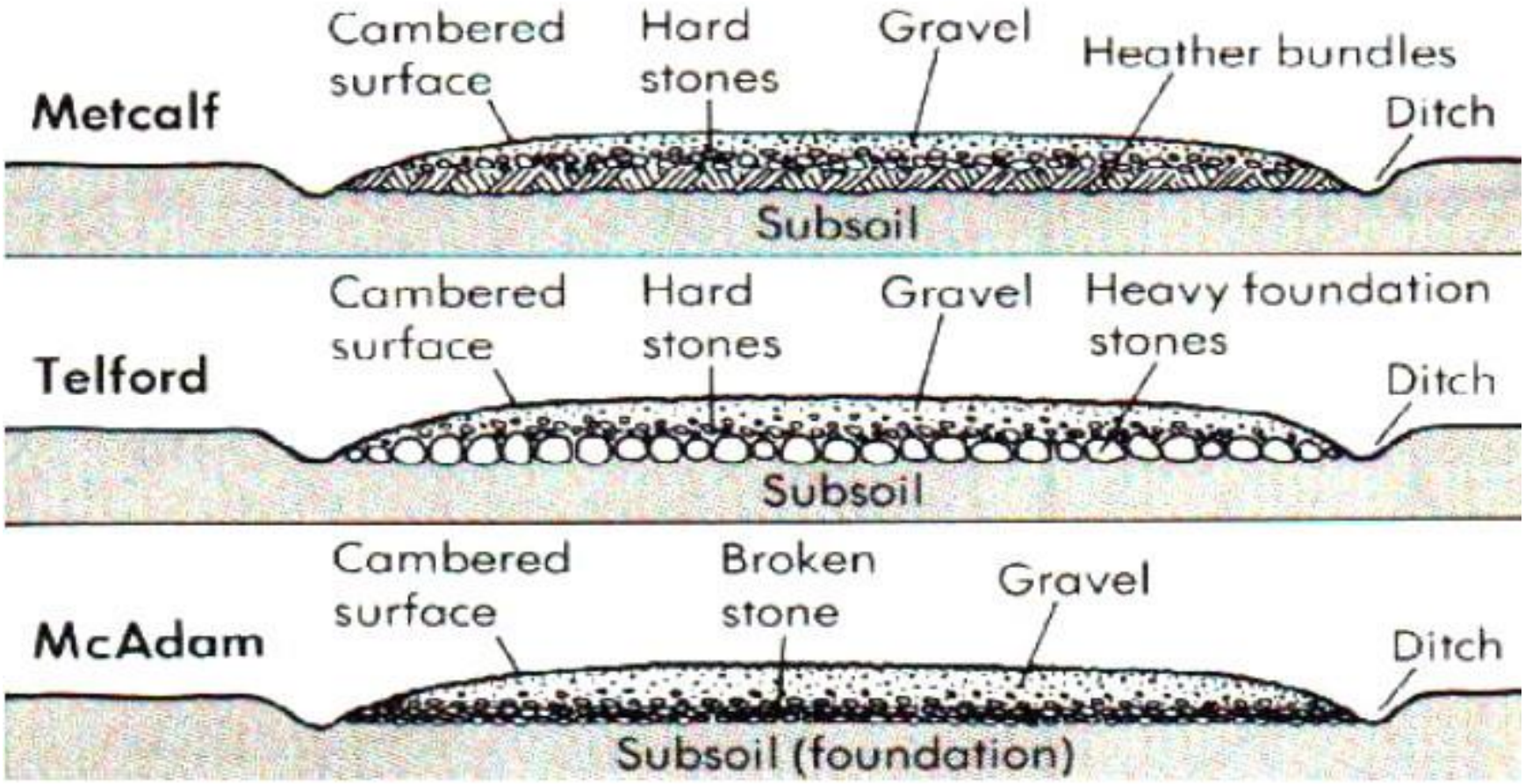
Tresaguet Method of Construction:

1) Pierre Tresaguet developed an improved method in **France** by the year **1764.**
2) **Subgrade** was prepared & a **layer of large foundation stones** were laid on edge by hand.
3) **At the edges** of the pavement **large stones** were embedded edgewise to serve as submerged kerb stones.
4) After that **broken stones** were **packed** to a thickness of about **8 cm** & compacted.
5) **Top wearing course** was made of **smaller stones** & compacted to a **thickness of 5 cm** & increasing towards center gradually with slope **1 in 45.**
6) **Thickness** of the road need be only **30 cm.**
7) Importance given to **subgrade moisture & drainage** of surface water.



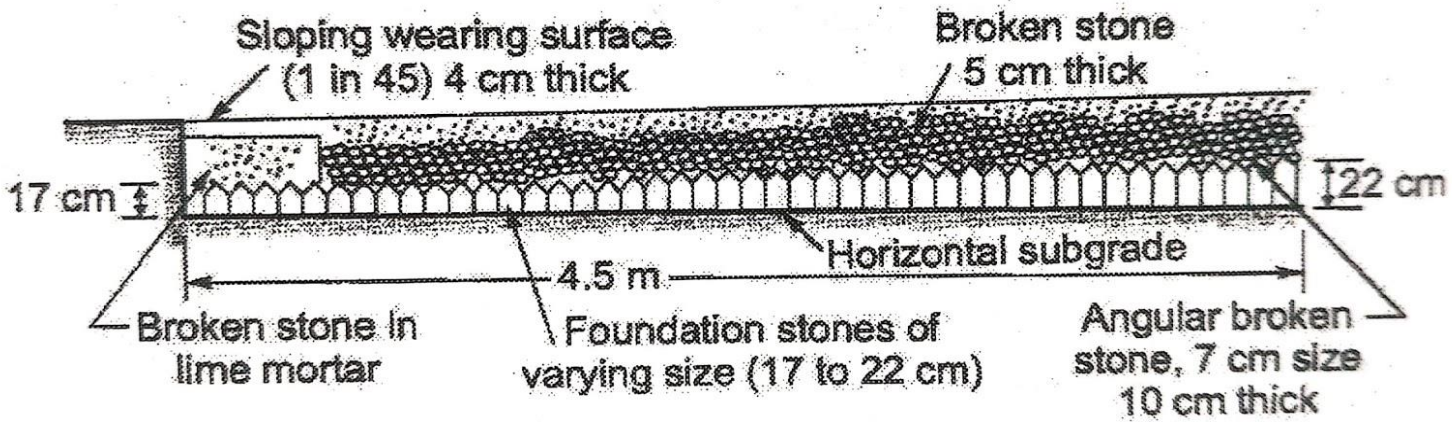
Metcalf Method:

1) When Tresaguet was working in France, John Metcalf **(1717-1810)** was working on road construction & development in **England.**
2) Good **Drainage System**.
3) He constructed about **290 km of road** in northern England.



Telford Method:

1) Thomas Telford was the founder of the **Institution of Civil Engineers at London**.
2) **1803**, A level **subgrade** was prepared to a **width of about 9m.**
3) **Large foundation stone** were laid with hand with largest face down so as to laid in stable position.
4) The interstices between foundation stones were filled with **smaller stone & chipping.**
5) The **central portion** of about **5.5 m width** was covered with **two layers of angular broken stones** to compacted thickness 10 & 5 cm.
6) **Lime mortar with broken stone** were also used to lateral stability towards edges of road.
7) Cross slope of about **1 in 45** was maintained.
8) Telford also provided **cross drains** at intervals of about **90 m.**



Macadam Method:

1) **1827, John Macadam** put forward an entirely **new method of road construction** as compared to all the previous methods.
2) The **soil subgrade** being **the lowest portion** of the pavement should be **prepared properly** & kept drained so as to carry the load transmitted through the pavement.
3) Therefore subgrade was compacted & prepared with a **cross slope of 1 in 36.**
4) Small size **broken stones were spread over** the prepared soil subgrade & compacted. Similarly next layer were constructed.
5) This technique helped to improve the **load dispersion characteristics** even though thickness were less than previous methods.
6) Water Bound Macadam, Penetration Macadam, Bituminous Macadam & Dense Bituminous Macadam method developed with slight modification .

Water Bound Macadam Method: WBM Method

1) **WBM** road construction method is **improved method** based on Macadam method.
2) **Crushed stones** mechanically **interlocked** by rolling & voids filled by screening & **binding material with water.**
3) Thickness of each **compacted layer** 7.5 to 10 cm.
4) **Coarse aggregate** of size 40 to 90 mm used for **sub-base.**
5) Subgrade i.e **lowest portion of pavement** is prepared with compacted layer of smaller size broken stones **carry the load transmitted**.
6) Compacted layer of smaller size stones at bottom **replaced the heavy foundation stones** used before this.
7) **Stresses** due to the wheel load of traffic **gets decreased** at the lower layer of pavement & larger size stones are not necessary at the lowest layer.
8) Compacted stones of **smaller size** are better for **load dispersion**.
9) Economical, locally available material is used & not required skilled labor.
10) Maintenance is **high & life** is less.


Tack Coat
Seal Coat
Prime Coat
Surface Course (25-50 mm)
Binder Course (50-100 mm)
Base Course (100-300 mm)
Subbase Course (100-300 mm)
Compacted Subgraded (150-300 mm)

WBM Method



Road Development in India:

1) Road were exist in **Mohenjo-Daro & Harappa**, 25 to 35 centuries (B.C.)

2) Emperor Ashoka had improved the roads.

3) Roads of India were also improved during the **Mughal periods.**

4) 1865, **Lord Dalhousie, Governor General formed PWD** & which constructed Grand Trunk Road.

For several centuries, it has **linked the eastern and western regions of the Indian subcontinent, running from Bengal, across north Republic of India, into Peshawar in Pakistan**

Progress of Road Development in India:

1)Road development in India started after initiation of the **Nagpur Road Plan in the year 1943.**

2)The road development works progressed with the implementation of **plans** with effect from the year **1951** & subsequent road development plan.

3)The **total road length** in the country has increased from **4 lakh km** in the year 1951 to **over 42 lakh km** .

4)Only 35-40% of total revenue from the road transport sector in the country has been **invested** on **road maintenance works & for development** plans during the past several decades.

Road Development in India: Jayakar Committee

1) 1927, the central government appointed the Jayakar Committee **to report on the condition of existing roads & to suggest ways to develop the road.**

2) **Central government** should **look** after all the important road.

3) The road development in the country **should be considered as a national interest.**

4) An **extra tax should be levied on petrol** from the road users to develop a road

5) Road development fund called **Central Road Fund.** It is formed by the year 1929.

6) The semi-official technical body called the **Indian Roads Congress (IRC)** was formed in 1934, to monitor design & construction of road.

7) A research organization should be instituted to **carry out research** and development work.

8) The **Central Road Research Institute** was started in 1950.

Indian Road Congress: IRC

1) Indian Road Congress is a **semi official technical body** established by central government **as per recommendation of Jayakar committee in 1934**.

2) IRC is formed **to monitor** the **standardization, specification, recommendation regarding design & construction** of roads.

3) It **provide a forum** for regular pooling of experience and ideas on all matters affecting the planning, construction and maintenance of roads in India.

4) IRC played important role in the **formulation of 20 year road development plans** in India.

5) Now IRC **has become successful an active body** of controlling specifications, standardization and recommendations on materials, design and construction of roads and bridges.

6) The IRC work with **Ministry of Road Transport** & Also publishes research journals.

Central Road Fund:

1) **Indian Legislature formed the Central Road Fund in 1929** as per recommendation of Jayakar Committee.

2) At that period the consumers of **petrol were charged an extra levy** of 2.64 paisa per litre.

3) Out of annual revenue collected, **20% was to be retained by central government for administration purpose**. **Balance 80%** was to be distributed for **construction & maintenance of road**.

Motor Vehicle Act:

1) In **1939 the motor vehicles Act was brought into effect** by Government of India to **regulate the road traffic in the form of traffic laws, ordinances and regulations.**

2) The three phases primarily covered are control of the **driver, vehicle ownership and vehicle operation** on roads and in traffic stream.

3) The **motor vehicle act** has been **appended** with several ordinances subsequently. The motor vehicle act has been **revised in the year 1988.**

Nagpur Road Congress:

1) A **conference** of the **chief Engineers** of all the states and provinces was conducted in 1943 by the Govt. of India at Nagpur, **at initiative of the IRC to finalize the first road development plan for the country** as a whole.

2) This is the **landmark in the history of road development in India**, as it was the **first attempt to prepare a co-ordinated road development programme** in a planned manner.

3) In this, **first 20 year road development plan** finalized & popularly known as the **Nagpur Road Plan.**

4) Target was to build **16 km length of road per 100 Km^2 area** of country.

National Highway Act:

1) **In 1956** the National Highway Act was passed.

2) The **responsibility of development** and maintenance of the National highway (NH) **to be provisionally taken by the central government.**

3) The central govt. to be empowered to declare any other highways as NH or to omit any of existing national highways from the list.

4) Also empowered to acquire land for the development of NH.

5) National Highway Authority of India (NHAI) started operating 1995.

Nagpur Road Plan (1943-63): First 20 Year Road Development Plan

1) **Responsibility of construction & maintenance of national highways** was assigned to the central government.
2) Nagpur Road Plan (1943-63) **aiming to provide** for **1,98,000 km of surface roads** and **3,33,600 km of un-surfaced roads** to reach 16 km road length per100 sq.km area.
3) The formulae were based on **star and grid pattern of road network**. But the existing irregular pattern of roads and obligatory points not fitting in the geometric pattern were to be given due consideration.
4) **Size of the grid** of road in **agricultural area** would be **16 km** so that the maximum distance from the center is 8 km and the average distance of the villages from metaled roads would be less than 3.2 km.
5) **In non-agricultural area** the **size of the grid is of 64 km sides**, the farthest distance from the center to the metaled roads being 32 km.
6) The second category roads are meant to provide internal road system linking small villages with first category roads.

Nagpur Road Plan: First 20 Year Road Development Plan

- The **length of railway tracks** in the area was also considered in deciding the length of first category road.
- **Recommendations were made for the geometric standards** of roads, bridge specifications & highway organizations.
- Two **formulae were finalized** at the Nagpur Conference **for deciding the road length** for the country as a whole as well as for individual areas (like district)
- The road network in the country was classified into **five categories**:

    1) National Highways,
    2) State Highways,
    3) Major District Road
    4) Other District Roads &
    5) Village Roads

Second 20 Year Road Development Plan: (1961-81) Bombay Road Plan

1) First 20 year road plan's **target road length** was nearly **completed earlier in 1961**, Hence next 20 year road plan commenced from 1961 by the IRC, sub committee and Chief Engineers.

2) The second road plan expected overall **road length of 10,57330 km** by the year 1981.The cost of the plan has been worked out to **Rs.5200 crore.**

3) The **target of this plan** is to provide a total road length of **32 km per 100 sq.km area** which is almost double of that achieved up to the year 1961.

4) Maximum distance of any place in a developed area would be **6.4 km from a metaled road** and 2.4 km from any category of roads.

5) The maximum distance from any place in as semi-developed area would be **12.8 km from a metaled road.**

6) Similarly the maximum distance in an undeveloped area would be **19.2 km from a metaled road** and 8 km from any road.

Second 20 Year Road Development Plan: (1961-81) Bombay Road Plan

7) Every town with population above **2000 in plains** and above **1000 in semi hill areas** and above **500 in hilly areas** should be connected by a metaled road.

8) Expressways 1600 km of length has been included in the proposed target of NH.

9) Length of railway track is **considered independent** of the road system and hence it is not subtracted to get the road length.

10)Five different formulae were framed to calculate the length of NH, SH, MDR, ODR & VR.

Third 20 Year Road Development Plan: (1981-2001) Lucknow Road Plan

1) **Villages** having **population over 500** should be connected by all weather roads.
2) Overall **density of road** in the country should be increased to **82 km per 100 sq. km.**
3) NH network should be expanded to form square grid of 100 km sides so that **no part** of the country is **more than 50 km away from a NH**.
4) All villages with population **over 1500** should be connected by **MDR** & population **1000 to 1500 by ODR.**
5) Villages of population **less than 500** should be connected by road **within a distance of 3 km** in plain & 5 km in hilly area.
6) Roads should be constructed in less industrialized area to **attract growth of industries.**
7) The **existing roads should be improved** by rectifying the defects in geometric design.
8) Improving the pavements surface & widening of pavements, as necessary, so that safety & economy is achieved.

National Highway Development Projects: 2000

1) **NHAI** took up the National Highway Development Projects by the year **2000.**

2) **Phase-I :** Golden Quadrilateral of total length 5864 km connecting 4 major cities**.**

    - **Delhi – Mumbai**
    - **Mumbai – Chennai**
    - **Chennai - Kolkatta**
    - **Kolkatta – Delhi**

3) **Phase-II :**

    - **North – South corridor** connecting Srinagar to Kanyakumari.
    - **East – West corridor** connecting Silchar to Porbandar.

4) **Phase-III, IV, V, VI & VII :** Development of NH, Expreesways & improvement of urban road network.

Classification of Roads:

Depending on whether they can be used during different season of the year:

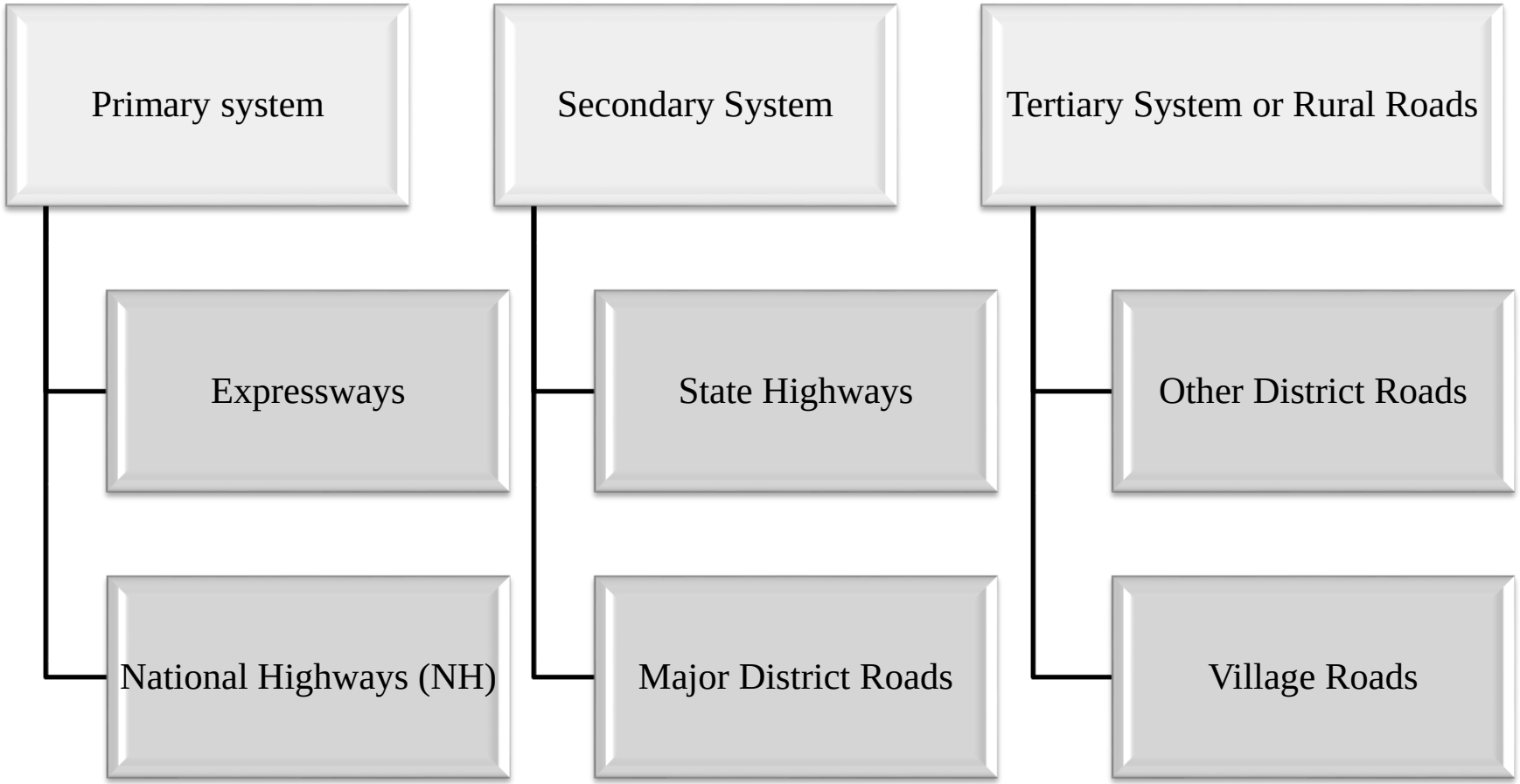
1) **All Weather Road:** negotiable during all season of the year.
2) **Fair Weather Road:** traffic may be interrupted during monsoon season.

Road Classification as per Nagpur Road Plan based on location & function:

1) **National Highway (NH):** connecting major ports, foreign highways, capitals of large states & industrial area. NH-4 Thane- Bangalore- Chennai.
2) **State Highway (SH):** Connecting NH of adjacent highways, district head quarter & important cities within state. Same design speed & geometric design specification used for NH & SH.
3) **Major District Roads (MDR):** Important roads within a district serving areas of production and markets. Also connecting with other major roads or SH.
4) **Other District Roads (ODR):** Roads are serving rural areas of production & markets.
5) **Village Roads (VR):** Connecting villages.

	1st 20 year road plan	2nd 20 year road plan	3rd 20 year road plan
Location	Nagpur	Bombay	Lucknow
Year	1943	1961	1981
Target	16km/100km²	32km/100km²	82km/100km²
Total road length	5.29lakhs km	10.57lakhs km	27.02lakhs km
Road Categories/ Addition	Roads were divided into 5 categories NH, SH, MDR, ODR, VR	Expressway was added	Roads were divided into 3 categories • Primary - NH & Express ways • Secondary - SH & MDR • ODR & VR

Modified Classification of Road System by Third Road Development Plan (1981-2001)



Road Patterns:

The various road pattern may be classified as follows:

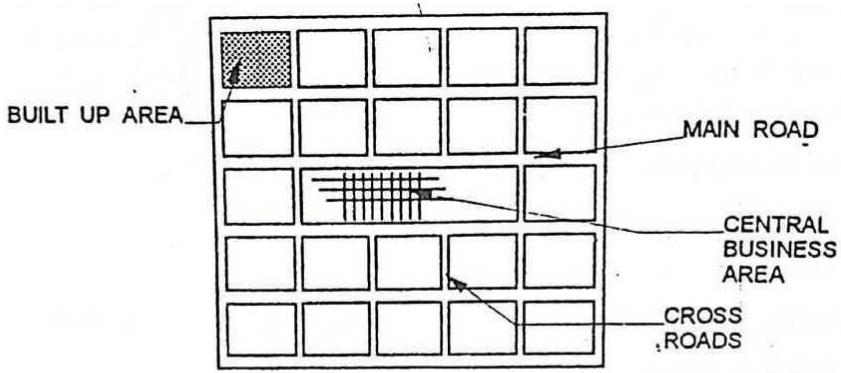
1) **Rectangular or Block Pattern:** Adopted in the city of roads of Chandigarh but from traffic operation point is not convenient

2) **Hexagonal Pattern:**

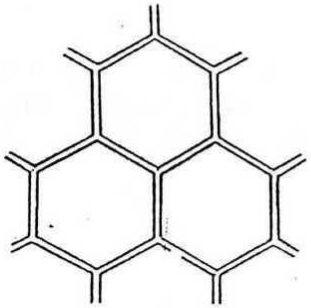
3) **Radial or Star & Block Pattern:**

4) **Radial or star & Circular Pattern:** Road network in Delhi

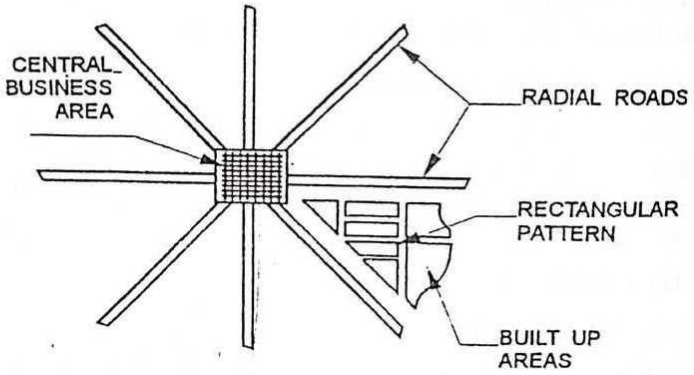
5) **Radial or Star & Grid Pattern:** Nagpur Road



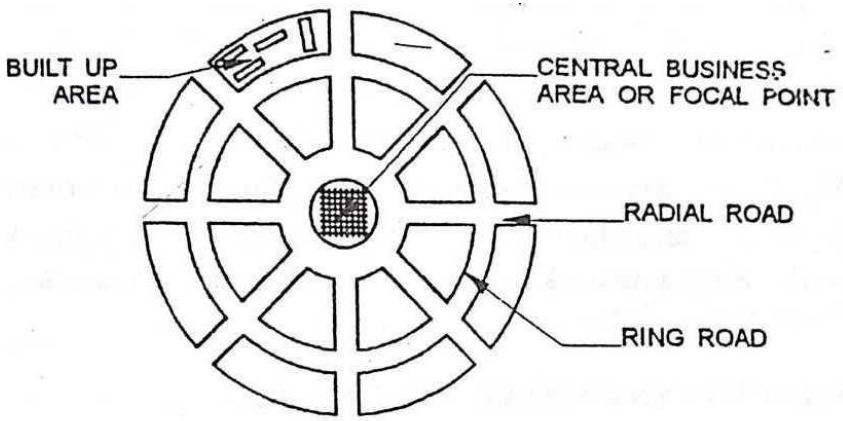
(a) Rectangular or block pattern



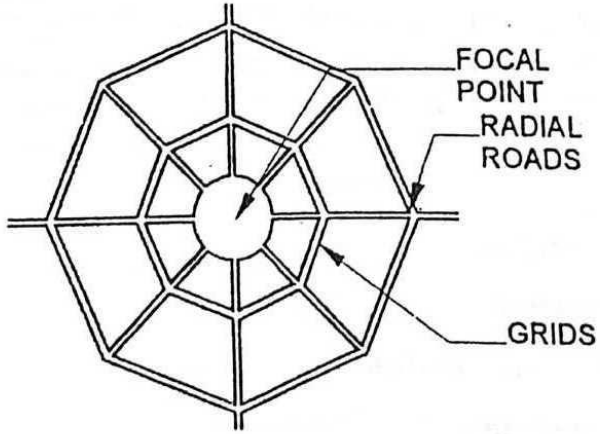
(b) Hexagonal pattern



(c) Radial or star and block pattern



(d) Radial or star and circular pattern



(e) Radial or star and grid pattern

Year/Period	Event/Development
2500-1500 B.C.	Roads existed in Mohenjo-Daro & Harappa.
1900 B.C.	First recorded road constructed by the Assyrian Empire.
312 B.C.	Romans pioneered large-scale road construction.
~273-232 B.C.	Emperor Ashoka improved roads in India.
Mughal Period	Roads in India further improved.
1764	Tresaguet Method of construction developed in France by Pierre Tresaguet.
1717-1810	John Metcalf developed roads and drainage systems in England, constructing about 290 km of roads.
1803	Thomas Telford introduced his road construction method in England.
1827	John Macadam developed a new road construction method, leading to Water Bound Macadam, Penetration Macadam, and Bituminous Macadam.
1865	Lord Dalhousie established PWD in India and constructed Grand Trunk Road.
1927	Jayakar Committee appointed to assess road conditions and suggest improvements.
1929	Central Road Fund established based on Jayakar Committee recommendations.
1934	Indian Roads Congress (IRC) formed to standardize road construction in India.
1939	Motor Vehicles Act introduced in India to regulate road traffic.
1943	Nagpur Road Plan (First 20-Year Road Development Plan) formulated.
1950	Central Road Research Institute established.
1951	Road development plans commenced in India, increasing road length from 4 lakh km to 42 lakh km over time.
1956	National Highway Act passed, making NH development a central government responsibility.
1961-1981	Second 20-Year Road Development Plan (Bombay Road Plan) initiated, aiming for a total road length of 10,57,330 km.
1981-2001	Third 20-Year Road Development Plan (Lucknow Road Plan) introduced, focusing on rural connectivity and road density improvements.
1995	National Highway Authority of India (NHAI) began operations.

All roads of the country classified into five categories:

① National Highway [NH] ② State Highway [SH]

③ Major District Roads [MDR] ④ Other District Roads [ODR]

⑤ Village Roads [VR]

Nagpur Road Plan assumes star and grid patterns of road network. Two plan formulae suggested to find the length of two categories of road.

Category-I : Surface or metalled roads [NH, SH & MDR]

Category-II : Un-surface roads [ODR & VR]

Nagpur road plan aimed at achieving a modest average road density of 16 km per km^2 area.

① The total length of the first category or surfaced roads for NH, SH & MDR in km is given by:

$$\text{Length of (NH+SH+MDR), km} = \left[\frac{A}{8} + \frac{B}{32} + 1.6N + 8T\right] + D - R$$

Where, A = Agricultural Area, km^2

B = Non-agricultural area, km^2

N = No. of villages with popn range 2001-5000.

T = No. of villages with popn over 5000.

D = Developpment allowance of 15% of road length calculated for agricultural & industrial development during next 20 year.

R = Existing length of railway track, km.

(II) The total length of second category roads for other district roads and village roads in kms is given by:

$$\text{Length of (ODR+VR), kms} = [0.32V + 0.8Q + 1.6P + 3.25] + D$$

where, V = No. of villages with popⁿ 500 or less.

Q = No. of villages with popⁿ range 501 – 1000.

P = No. of villages with popⁿ range 1001 – 2000

S = No. of villages with popⁿ range 2001 – 5000.

D = Developement allowance of 15% for next 20 years.

From the two formulae, it may be seen that in addition to the road length based on agricultural & non-agricultural areas, specific road length were allocated for villages of different population ranges.

Que. ① The following data were collected for planning the road development programme of a backward district.

ⓐ Total Area = 9600 km²
ⓑ Agricultural & Developed Area = 3200 km²
ⓒ Existing Railway Track Length = 105 km.
ⓓ Existing Length of Surface Road = 322 km
ⓔ Existing Length of Unsurface Road = 450 km
ⓕ Number of villages in different population range:

Population	>5000	2001-5000	1001-2000	501-1000	<500
No. of Villages	8	40	130	280	590

Calculate the additional lengths of surfaced and un-surfaced roads for the road system based on Nagpur Road Plan formulae for this district?

Solution :- D = 15% of Total Length of Road.

Ⓘ Total Length of Surfaced Road in km by Category - I Formula :-

$$= \frac{A}{8} + \frac{B}{32} + 1.6N + 8T + D - R$$

$$= \left[\left[\frac{3200}{8} + \frac{(9600-3200)}{32} + (1.6\times 40) + (8\times 8)\right] \times \frac{115}{100}\right] - 105$$

$$= 732.2 \text{ km}$$

Additional Length of Surfaced Road Required in km

$$= 732.2 - 322$$

$$= 410.2 \text{ km}$$

(II) Total Length of Unsurfaced Road in km by Category-II

formula :—

$$= [0.32V + 0.8Q + 1.6P + 3.2S] + D$$

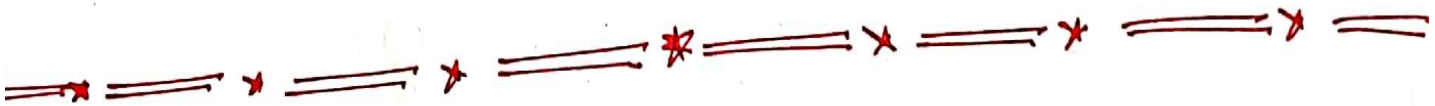
D ⇒ 15% of Road Length.

$$\therefore \quad = \left[(0.32 \times 590) + (0.8 \times 280) + (1.6 \times 130) + (3.2 \times 40)\right] \times \frac{115}{100}$$

$$= 861 \text{ km}$$

Additional Unsurfaced Road Required = 861 - 450

= 411 km



Second Twenty Year Road Plan :-

Bombay road plan aimed to achieve an average road density of 32 km per km^2 area of country by 1981.

Five formulae were framed to calculate length of NH, SH, MDR, ODR and VR.

① Length of NH, km $= \left[\frac{A}{64} + \frac{B}{80} + \frac{C}{96}\right] + [32k + 8M] + D$

② Length of NH & SH in km $= \left[\frac{A}{20} + \frac{B}{24} + \frac{C}{32}\right] + [48k + 24M + 11.2N + 6P]$

③ Length of NH, SH & MDR, in km ⇒

$= \left[\frac{A}{8} + \frac{B}{16} + \frac{C}{24}\right] + [48k + 24M + 11.2N + 9.6P + 6.4Q + 2.4R] + D$

④ Length of NH, SH, MDR & ODR, in km ⇒

$= \left[\frac{3A}{16} + \frac{3B}{32} + \frac{C}{16}\right] + [48k + 24M + 11.2N + 9.6P + 12.8Q + 4R + 0.8S + 0.32T] + D$

⑤ Length of NH, SH, MDR, ODR & VR, in km ⇒ Total Length of All Roads

$= \left[\frac{A}{4} + \frac{B}{8} + \frac{C}{12}\right] + [48k + 24M + 11.2N + 9.6P + 12.8Q + 5.9R + 1.6S + 0.64T + 0.2V] + D$

A = Developed and agricultural areas, km^2

B = Semi-developed area, km^2

C = Undeveloped area, km^2

K = Number of towns with population over 1,00,000

M = No. of towns with pop^n range 50,000 – 1,00,000.

N = No. of towns with pop^n range 20,000 – 50,000.

P = No. of towns with pop^n range 10,000 – 20,000.

Q = No. of towns with pop^n range 5,000 – 10,000.

R = No. of towns with pop^n range 2000 – 5000.

S = No. of towns with pop^n range 1000 – 2000.

T = No. of towns with pop^n range 500 – 1000.

V = No. of towns with range below 500.

D = Development allowance of 5% of $\frac{\text{Road}}{\text{---}}$ Length calculated for further development & other unforeseen factors.

Que. ② Calculate the total lengths of NH, SH, MDR, ODR & VR needed in a district as per second 20-year road development plan for following data:

Total Area = 18400 km^2

Developed & Agricultural Area = 8000 km^2

Undeveloped Area = 4800 km^2

Population Range	Number of Towns
< 500	200
500 – 1000	350
1000 – 2000	750
2000 – 5000	360
5000 – 10000	150
10000 – 20000	80
20000 – 50000	25
50000 – 100000	10
> 100 000	5

Sol^n :– B = Semi developed Area, km^2 ...

B = Total Area – [Developed Area + Undeveloped Area]

B = 18400 – (8000 + 4800)

$\boxed{B = 5600\ km^2}$

D = 5 %

(I) Length of NH, km $= \frac{A}{64} + \frac{B}{80} + \frac{C}{96} + (32k + 8M) + D$

$= \left[\frac{8000}{64} + \frac{5600}{80} + \frac{4800}{96} + (32\times5) + (8\times10)\right] + 5\%\text{ of RL}$

$= [125 + 70 + 50 + 160 + 80] \times \frac{105}{100}$

NH = 509.25 km

(II) Length of (NH + SH), km $= \left[\frac{A}{20} + \frac{B}{24} + \frac{C}{32} + 48k + 24M + 11.2N + 1.6P\right] + D$

$= \left[\frac{8000}{20} + \frac{5600}{24} + \frac{4800}{32} + (48\times5) + (24\times10) + (11.2\times25) + (1.6\times80)\right] + 5\%\text{ of RL}$

$= [400 + 233 + 150 + 240 + 240 + 280 + 128] \frac{105}{100}$

$= \underline{1754.5\text{ km}}$ $\quad \therefore$ SH = 1754.5 − 509.25

SH = 1245.25 km

(III) Length of (NH + SH + MDR), km :—

$= \left[\frac{A}{8} + \frac{B}{16} + \frac{C}{24}\right] + [48k + 24M + 11.2N + 9.6P + 6.4Q + 2.4R] + D$

$= \left[\frac{8000}{8} + \frac{5600}{16} + \frac{4800}{24} + (48\times5) + (24\times10) + (11.2\times25) + (9.6\times80) + (6.4\times150) + (2.4\times360)\right] + 5\%\text{ of RL}$

$= [1000 + 350 + 200 + 240 + 240 + 280 + 768 + 960 + 864] \frac{105}{100}$

$= \underline{5147\text{ km}}$

MDR = 5147 − 1754.5

$\therefore$ **MDR in km = 3392.5 km**

(IV) Length of NH, SH, MDR, & ODR in km :—

$$= \left[\frac{3A}{16} + \frac{3B}{32} + \frac{C}{16}\right] + \left[48K + 24M + 11.2N + 9.6P + 12.8Q + 4R + 0.8S + 0.32T\right] + D$$

$$= \left[\frac{3\times8000}{16} * \frac{3\times5600}{32} + \frac{4800}{16} + (48\times5) + (24\times10) + (11.2\times25) + (9.6\times80) + (12.8\times150) + (4\times360) + (0.8\times750) + (0.32\times350)\right] + 5\% \text{ of RL}$$

$$= 7925 \times \frac{105}{100}$$

$$= 8321.2 \text{ km}$$

$\therefore$ ODR = 8321.2 − 5147

ODR = 3174.2 km

(V) Length of NH, SH, MDR, ODR & VR in km :—

$$= \left[\frac{A}{4} + \frac{B}{8} + \frac{C}{12}\right] + \left[48K + 24M + 11.2N + 9.6P + 12.8Q + 5.9R + 1.6S + 0.64T + 0.2V\right] + D$$

$$= \left[\left(\frac{8000}{4}\right) + \left(\frac{5600}{8}\right) + \left(\frac{4800}{12}\right) + (48\times5) + (24\times10) + (11.2\times25) + (9.6\times80) + (12.8\times150) + (5.9\times360) + (1.6\times750) + (0.64\times350) + (0.2\times200)\right] + 5\% \text{ of RL}$$

$$= \left[2000 + 700 + 400 + 240 + 240 + 286 + 768 + 1920 + 2124 + 1200 + 224 + 40\right] \times \frac{105}{100}$$

$$= \underline{10642.8 \text{ km}}$$

$\therefore$ VR in km = 10642.8 − 8321.2

$\therefore$ **VR = 2321.6 km**

The Roads Lengths Required are as below:-

NH = 509.25 km

SH = 1245.25 km

MDR = 3392.5 km

ODR = 3174.2 km

VR = 2321.6 km

Road Length by Third 20 Year Road Development Plan :-

(I) Primary Road System :-

Total Length of NH in the country or state can be obtained by dividing the total area of country or state by 50.

$$\text{Length of NH} = \frac{\text{Area}}{50}$$

(II) Secondary Road System :-

(a) Total Length of SH in the Country or State could be obtained

$$= [62.5 \times \text{No. of Towns}] - [\text{Length of NH}]$$

(b) Total Length of SH in the country could be obtained by

$$= \frac{\text{Area}}{25}$$

Note :- Adopt the length of SH, whichever is more.

(c) Total Length of MDR in the country could be obtained by

$$= \frac{\text{Area}}{12.5}$$

(d) Total Length of MDR in the country could be obtained by

$$= 90 \times \text{No. of Towns}$$

Note :- Provide the length of MDR (Higher of the Two)

(III) Tertiary Road System :-

Total Length of All categories of Road $= (\text{Area}) \times \frac{82}{100}$

$$(NH + SH + MDR + ODR + VR) = (\text{Area of Country}) \times \frac{82}{100}$$

Que. ③ The area of a certain district in India is 13400 km² and there are 12 towns as per 1981 census. Determine the lengths of different categories of roads to be provided in this district by the year 2001

Solution :—

ⓐ Length of NH in km = $\frac{\text{Area}}{50} \Rightarrow \frac{13400}{50} \Rightarrow$ **268 km**

ⓑ Length of SH in km = $\frac{\text{Area}}{25} \Rightarrow \frac{13400}{25} \Rightarrow$ 536 km

ⓒ Length of SH in km = (62.5 × No. of Towns) − Length of NH

= (62.5 × 12) − 168

= 482 km

Adopt Higher of ⓑ & ⓒ i.e. **Length of SH = 536 km**

ⓓ Length of MDR = $\frac{\text{Area}}{12.5} \Rightarrow \frac{13400}{12.5} \Rightarrow$ 1072 km

ⓔ Length of MDR = (90 × No. of Towns)

= 90 × 12

= 1080 km

Provide Higher of ⓓ & ⓔ i.e. **Length of MDR = 1080 km**

ⓕ Total Length of All Category Road = (Area) × $\frac{82}{100}$

(NH+SH+MDR+ODR+VR) = 13400 × $\frac{82}{100}$

NH+SH+MDR+ODR+VR = 10988 km

But NH+SH+MDR ⇒ 268 + 536 + 1080 ⇒ 1884

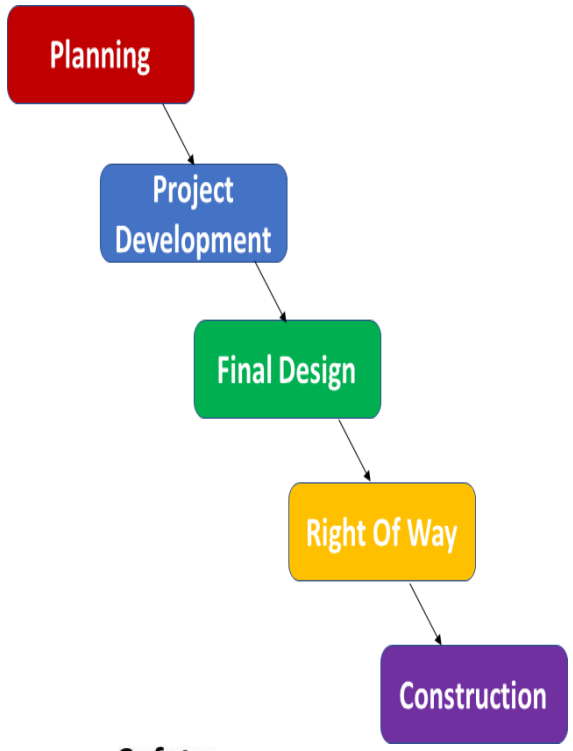
∴ ODR+VR = 10988 − 1884

ODR+VR = 9104 km

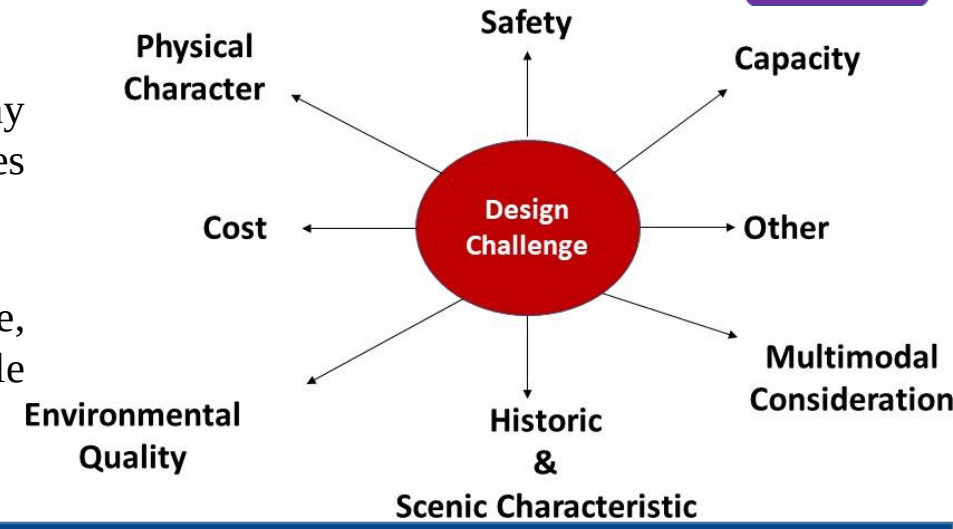
Highway Planning

The problems identified usually fall into one or more of the following four categories:

- The existing physical structure needs major repair/replacement **(structure repair)**
- Existing or projected future travel demands exceed available capacity, and access to transportation and mobility need to increased **(capacity).**
- The route is experiencing an inordinate number of safety and accident problems that can only resolved through physical, geometric changes **(safety).**
- Developmental pressures along the route make a reexamination of the number, location, and physical design of access points necessary **(access).**



- The initial definition of the need for any highway or bridge improvement project takes place during the planning stage.
- This problem definition occurs at the State, regional, or local level, depending on the scale of the proposed improvement.



Highway Planning

Project Development

- After a project has planned and programmed for implementation, it moves into the **project development phase.** At this stage, the environmental analysis intensifies.
- The level of environmental review varies widely, depending on the **scale and impact** of the project.
- It can range from a multiyear effort to prepare an **Environmental Impact Statement** (a comprehensive document that analyzes the potential impact of proposed alternatives) to a modest environmental review completed in a matter of weeks.

Final Design

- After a preferred alternative has selected and the project description agreed upon as stated in the **environmental document**, a project can move into the final design stage.
- Depending on the **scale and complexity** of the project, the final design process may take from a few months to several years.
- The following paragraphs discuss some important considerations of design, including:
  - ✓ Developing a concept
  - ✓ Considering scale and
  - ✓ Detailing the design

Highway Planning

Developing a Concept

- A design concept gives the project a focus and helps to move it toward a specific direction.
- Some of the many elements of highway design are,
  - ✓ Number and width of travel lanes, median type and shoulders
  - ✓ Traffic barriers
  - ✓ Overpasses/bridges
  - ✓ Horizontal and vertical alignment and affiliated landscape.

Considering Scale

- People driving in a car see the world at a much different scale than people walking on the street.
- This large discrepancy in the design scale for a car versus the design scale for people has changed the overall planning of our communities.
- The design element with the greatest effect on the scale of the roadway is its width, or cross section.

Detailing The Design

- Particularly during the final design phase, it is the details associated with the project that are important.
- Employing a multidisciplinary design team ensures that important design details considered and those they are compatible with community values.
- Often it is the details of the project that are most recognizable to the public.

Highway Planning

Right-of-way, Construction, And Maintenance

- Once the final designs have prepared and needed right-of-way is purchased, construction bid packages made available, a contractor selected, and construction initiated.
- During the right-of-way acquisition and construction stages, minor adjustments in the design may be necessary; therefore, there should be continuous involvement of the design team throughout these stages.
- Construction may be simple or complex and may require a few months to several years.
- Once construction has completed, the facility is ready to begin its normal sequence of operations and maintenance.
- Even after the completion of construction, the character of a road can changed by inappropriate maintenance actions.
- For example, the replacement of sections of guardrail damaged or destroyed in crashes commonly utilizes whatever spare guardrail sections may be available to the local highway maintenance personnel at the time.

Alignment and Surveys

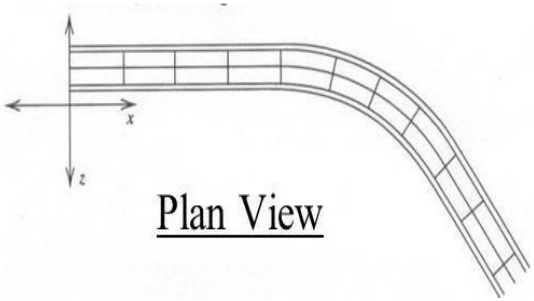
- The position of the **center line** on the highway in the ground is called highway alignment.
- Highway alignment includes horizontal alignment and vertical alignment.
- The projection of highway alignment in the **horizontal plane** is termed as the horizontal alignment and the projection of highway alignment in the **vertical plane** is called vertical alignment.
- Alignment must be selected in such a way that the overall **cost during construction, operation and maintenance** is minimum.
- Road design outputs are in the form of following drawings:

✓ **Plan**: Includes centre line, structures, Right of Way (ROW), carriage way, shoulders, side drain.

✓ **Longitudinal Profile**: Soil Type, Depth of cut, Height of Fill, Side drain (Information on from which chainage to which chainage), Direction of flow in the drain.

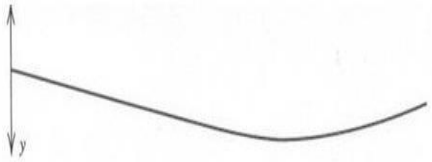
✓ **Cross section**: Ground Level, Formation Level, Superelevation, Area of Cutting and Area of Filling thus computation of the volume and then cost estimation can be done.

Horizontal Alignment



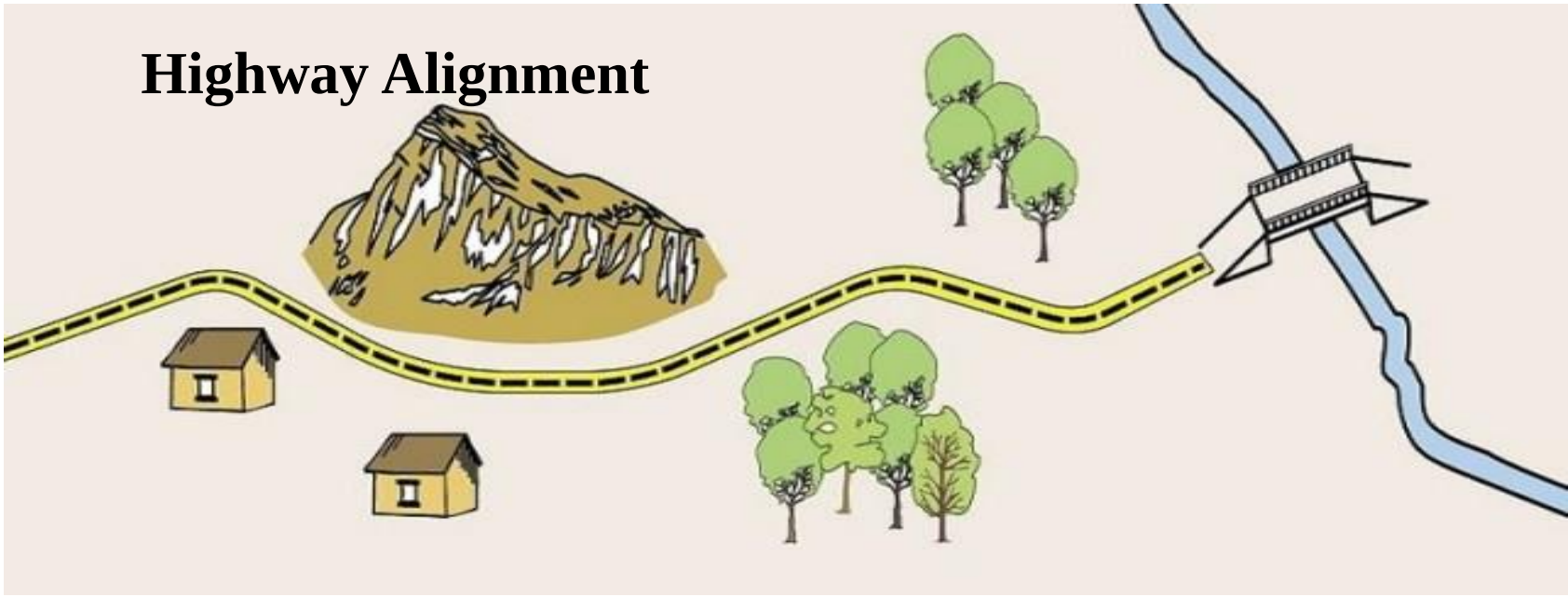
Plan View

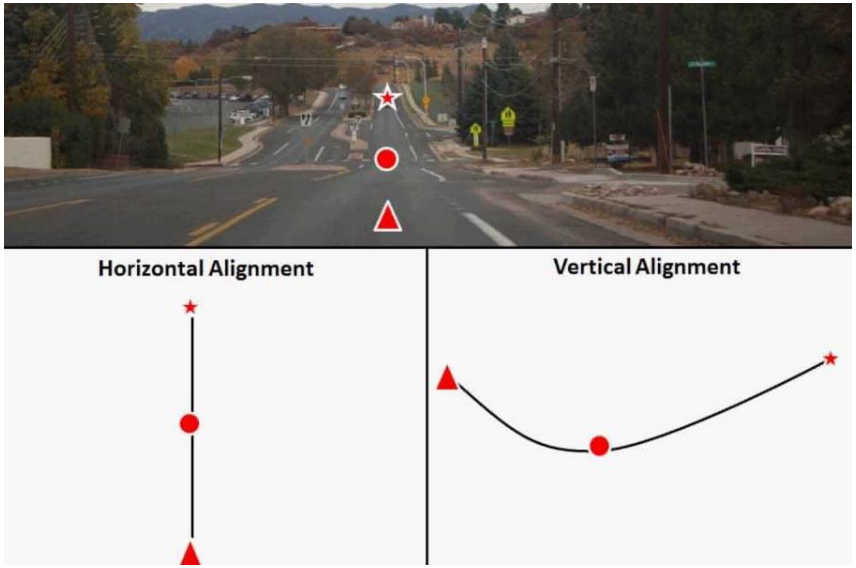
Vertical Alignment



Profile View

Highway Alignment







REQUIREMENTS OF HIGHWAY ALIGNMENT



The ideal alignment must have the following requirements:

- Safe (S)
- Easy (E)
- Short (S)
- Economical (E)
- Comfort (C)

The requirements can be memorized as **SESEC**.

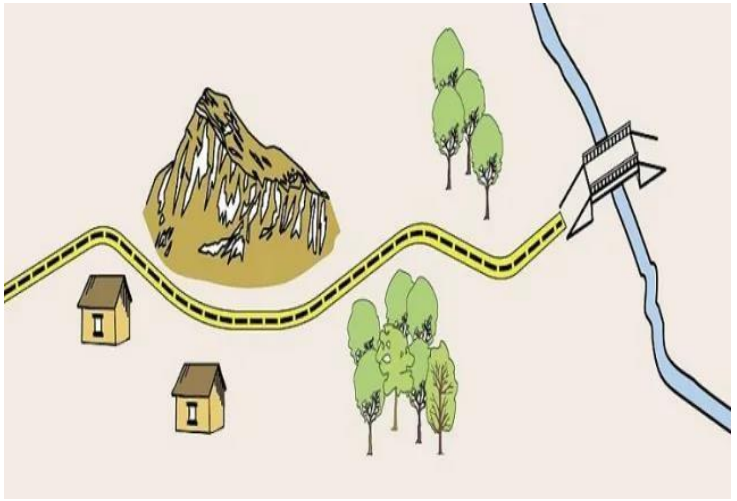
- **Safe:** The alignment need to be safe during construction, operation and maintenance especially at slopes, embankments and cutting.
- **Easy:** The construction materials if present at the place of construction makes the construction easier. Similarly, it should be easy during the operation of vehicles with easy gradients and curves.
- **Short:** The distance between the initial and final point need to be short so as to reduce the construction cost.
- **Economical:** The alignment should be economical during construction, operation, and maintenance. However, if the construction turned out to be economical, the gradient may not be easy which in turns increases the cost of operation and maintenance..
- **Comfort:** The alignment should be fixed such that it provides comfort to the drivers and the passengers.

Factors Controlling Highway Alignment:

1. Government Alignment:

As the road project needs a large investment, the government should be clear about the requirement of the road (when, what, how and why to construct).

2. Obligatory Points: obligatory points through which the road **alignment has to pass** are generally due to the topographic & other site condition including natural obstruction. **Mountain pass, Bridge to cross a river**, etc. There are obligatory points through which the road should **not pass** like **religious places, costly structure, unsuitable land,** etc.



Positive Obligatory Points: These are those points through which the alignment should pass.

- Existing Road
- Intermediate Town

Negative Obligatory Points: These are those points through which the alignment should not pass.

- Valleys, ponds, and marshy land
- Religious places
- Costly structures present in the way of alignment
- Conservation areas and restricted zones.
- Densely populated area.
- The road should not be within the boundary of the country.

3. Traffic: Road alignment should be decided based on the **requirements of road traffic. Traffic flow, Traffic volume, Future trends, etc.**

4.Geometric Design: Gradient, radius of curve, & sight distance also would govern the final alignment of road..

5.Economics: While trying to decreasing the **initial construction cost**, either the future road **maintenance cost** or **vehicle operation cost** or both of these may increase considerably. Therefore, economic analysis should be done properly based on life cycle cost.

6. Other Consideration: others factors such as **horizontal, vertical alignment, drainage consideration, hydrological factors, political consideration & monotony.**

SPECIAL CONSIDERATION IN HILL ROADS:

- Stability
- Drainage
- Geometric Standards
- Resisting Length

- The alignment needs to be fixed such that the number of **cross drainage structures** are less.

- Setting the straight alignment leads to **monotonous** driving.

Factors controlling highway alignment in India

Category	Key Factors
Topography	- Avoid steep slopes and difficult terrains - Minimize earthwork (cutting & filling) - Consider valley vs. ridge alignment in hilly areas
Geological & Soil Conditions	- Avoid landslide-prone and weak soil areas - Ensure foundation stability for bridges & embankments - Choose stable terrain to minimize future maintenance
Climate & Weather Conditions	- Consider heavy rainfall areas for proper drainage - Snowfall-prone regions require frost-resistant materials - High-temperature zones need heat-resistant road materials
Traffic Flow & Road Connectivity	- Ensure smooth integration with existing roads - Provide direct & shortest possible route - Consider future expansion needs
Drainage & Hydrology	- Proper cross-drainage structures (culverts, bridges) - Avoid flood-prone areas & ensure efficient water disposal - Design embankments to withstand water logging
Land Use & Social Considerations	- Minimize displacement of people & property - Avoid cutting through dense residential/commercial areas - Ensure accessibility to important locations (cities, industries, etc.)
Economic Factors	- Balance between initial cost & long-term maintenance - Use cost-effective design while maintaining quality - Consider land acquisition costs
Environmental & Ecological Concerns	- Minimize deforestation and protect wildlife areas - Control pollution and noise levels near sensitive zones - Ensure compliance with environmental regulations
Defence & Strategic Importance	- Ensure connectivity to border areas for military access - Consider security aspects in sensitive regions
Government Policies & Legal Regulations	- Follow IRC (Indian Road Congress) guidelines - Comply with National Highway Authority of India (NHAI) standards - Obtain necessary land and environmental clearances

ENGINEERING SURVEY AND ITS STAGES

- **Map Study**
  - ✓ Alignment avoiding valley, ponds, lakes.
  - ✓ When the road has to carry a row of hills, mountain pass may be the suitable alternative.
  - ✓ Approximate location of the bridge site.
- **Reconnaissance**
  - ✓ Valley, pond, lakes and other features that were not present in the topographical map.
  - ✓ A number of cross drainage structures, High Flood Level (HFL), Natural Ground Level.
  - ✓ Values of the gradient, the length of gradients and radius of the circular curve.
  - ✓ Soil type along the routes from field identification tests and observation of the geological features.
  - ✓ Sources of construction materials.
- **Preliminary Survey**
  - ✓ To collect necessary physical information and details of topography, drainage, and soil.
  - ✓ To compare different proposal in view of the requirement of good alignment.
  - ✓ To estimate the quantity of earthwork.
  - ✓ To finalize the best alignment.
- **Final Location and Detailed Survey**
  - ✓ The centre line of the road which is finalized in the preliminary survey is then located in the field by establishing the centre line.
  - ✓ The data during the detailed survey should be elaborated and completed for the preparation of the plans, designing, and estimation of the project.

Survey to be conducted for finalizing the Alignment:

1) **Topographic Map Study:** The process of making several possible alternative alignments of a road on the topographical map of the area is known as map study.

2) **Reconnaissance Survey:** A general examination of a fairly broad stretch of land between the terminal stations in the field, along the proposed alternative alignment marked on the map is known as reconnaissance survey. Engineer visit the site & examine the general characteristics of area. Rapid & rough survey between terminal station.

3) **Preliminary Survey:** The art of finding the details of alternative alignment found suitable during the reconnaissance survey is known as preliminary survey. Field to office work.

4) **Detailed Survey or Location Survey:** The detailed examination of the field along the alignment finally recommend during the preliminary survey is called as location survey.

Surveys to be conducted for finalizing the alignment of a road in India

Type of Survey	Purpose	Key Considerations
Reconnaissance Survey	Preliminary assessment of possible routes	- Identify feasible alignments - Observe terrain, obstacles, and natural features - Collect rough cost estimates
Topographical Survey	Detailed mapping of land features	- Measure elevations, slopes, and contours - Identify rivers, valleys, and ridges - Assist in route selection based on terrain
Geological Survey	Study of soil and rock conditions	- Assess stability of hills and slopes - Identify landslide-prone zones - Determine suitable foundation locations
Soil Investigation & Subsurface Survey	Determine soil properties and strength	- Identify load-bearing capacity of soil - Classify soil type (clay, sand, gravel, etc.) - Plan for suitable pavement design
Hydrological Survey	Study of drainage and water flow	- Assess flood-prone areas - Plan culverts, bridges, and drainage structures - Ensure proper water disposal to prevent road damage
Traffic Survey	Analyze current and future traffic demand	- Study vehicle types and volume - Consider peak-hour congestion - Plan for road capacity and future expansion
Environmental Survey	Assess ecological and environmental impact	- Minimize deforestation and wildlife disturbance - Reduce air, noise, and water pollution - Ensure compliance with environmental regulations
Land Acquisition & Social Impact Survey	Evaluate land ownership and resettlement needs	- Identify affected properties and communities - Assess compensation and rehabilitation requirements - Minimize displacement and social impact
Economic & Cost Survey	Estimate construction and maintenance costs	- Compare cost of alternative alignments - Optimize alignment for cost-effectiveness - Plan budget allocation for the project